

Software Engineering with Generative AI

Week 1&2 · Language Models & Transformers
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CS 5090 · Fall 2026
Michigan Technological University



Definition of Large Language Model

- Language Model: "A language model is a statistical or machine learning model that assigns probabilities to sequences of tokens."
- How "large"?
 - Model size: tens (or hundreds) of billions of parameters
 - Training data: 100s of billions of tokens or more
 - Compute: a lot of GPUs

A closer look of GPT-3

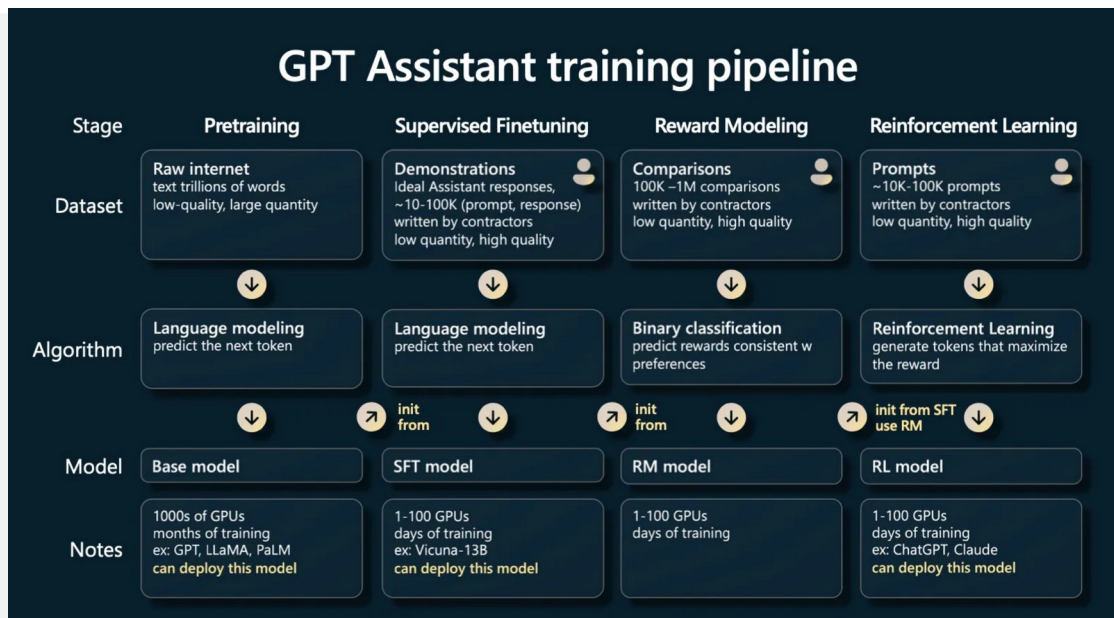
Model Name ↕	n_{params} ↕
GPT-3 Small	125M
GPT-3 Medium	350M
GPT-3 Large	760M
GPT-3 XL	1.3B
GPT-3 2.7B	2.7B
GPT-3 6.7B	6.7B
GPT-3 13B	13.0B
GPT-3 175B	175.0B

GPT-3 training data^{[1]:9}

Dataset	# tokens	Proportion within training
Common Crawl	410 billion	60%
WebText2	19 billion	22%
Books1	12 billion	8%
Books2	55 billion	8%
Wikipedia	3 billion	3%

Source: wikipedia

GPT training pipeline

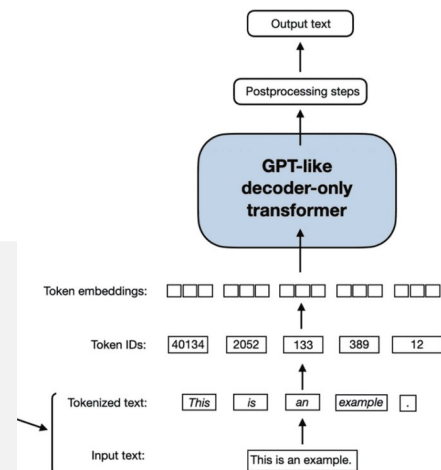
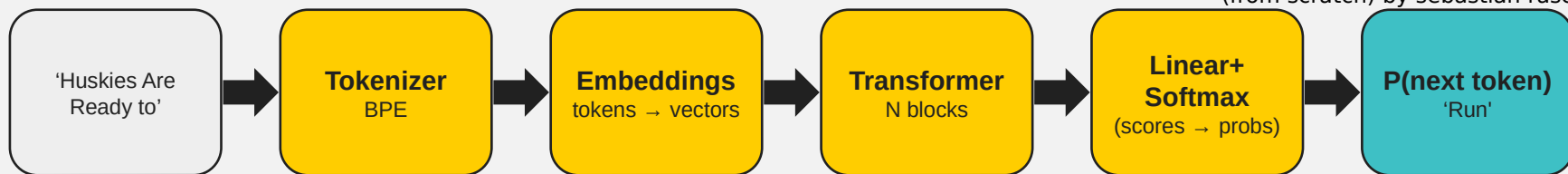


Source:

<https://medium.com/@chassweeting/the-state-of-gpt-by-andrew-kapathy-fad2f007c1b9>

From text to prediction

The whole pipeline this week, in one row



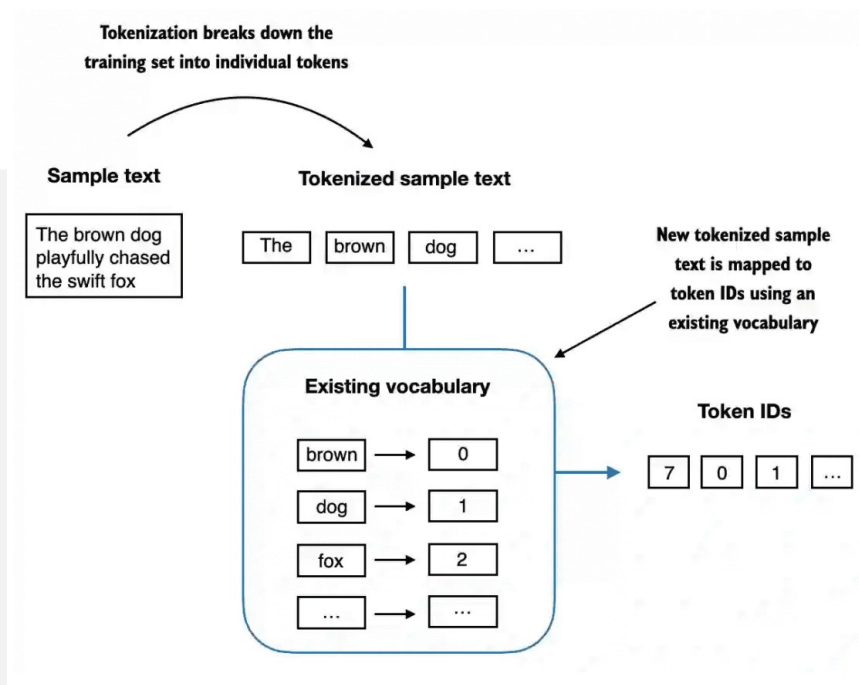
Source: build a large language model (from scratch) by sebastian raschka

Tokenization

● Methods:

- Word tokenization
- Subword tokenization
- Character-level tokenization

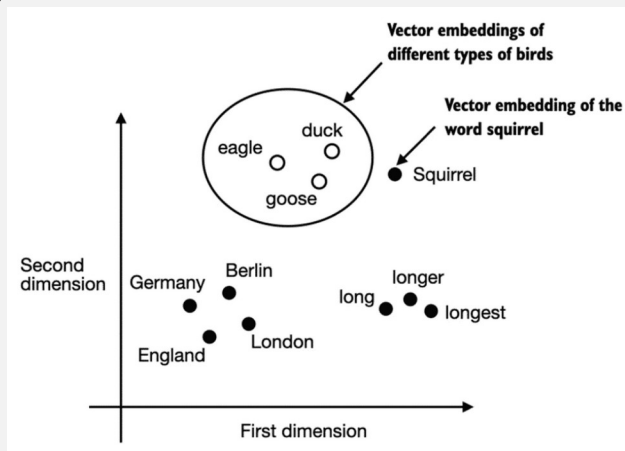
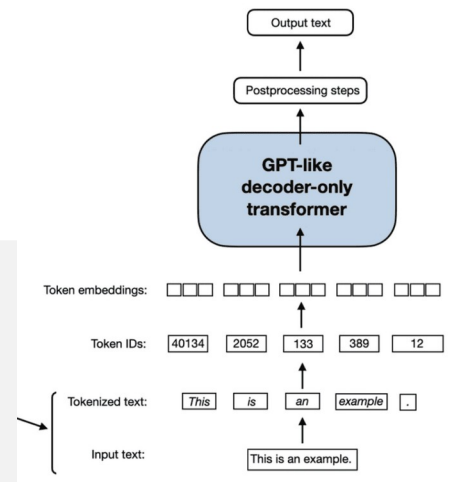
Method	Pros	Cons
Word-level	• Simple • Interpretable	• Risk of OOV • Does not leverage knowledge of root
Subword-level e.g. WordPiece, BPE	• Leverages common prefixes and suffixes • Learned from the data	• Risk of OOV, though less than word-level
Character-level	• Small chance of OOV • RoBUsT to CASinG anD MIspellINGS	• Makes computations slower • Embeddings not interpretable



Source: Transformers & LLMs cheatsheet,
Build a large language model (from scratch) by sebastian raschka

Word Embeddings

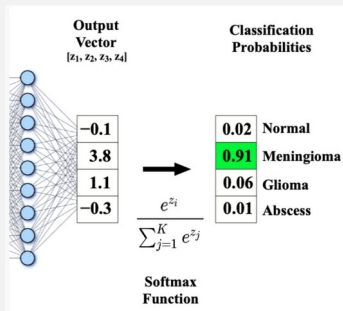
- We need a way to represent words as continuous-valued vectors for neural networks
- Parameters are also trained
- Why is there semantic structure?



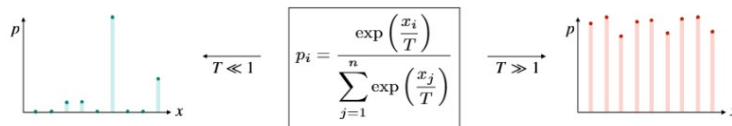
Source: Build a large language model (from scratch) by sebastian raschka

What comes after transformer blocks? Next token prediction

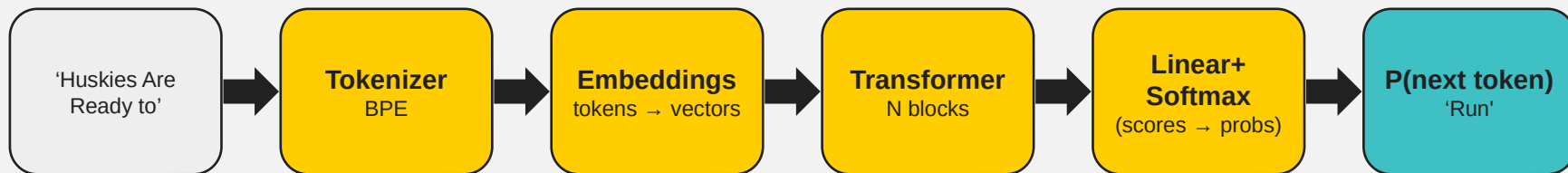
- Size=d-model $\rightarrow$ Linear $\rightarrow$ Size=|V|
- Softmax



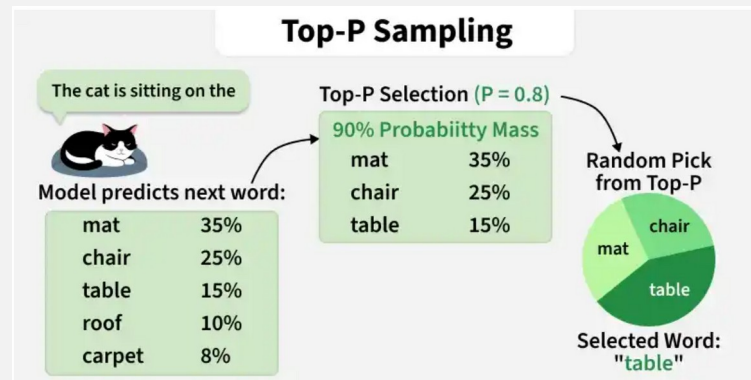
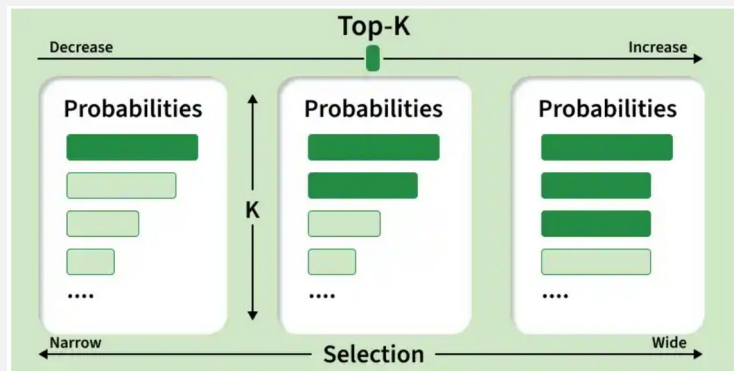
□ **Decoding sampling** – Token predictions are sampled from the predicted probability distribution p_i , which is controlled by the hyperparameter temperature T .



Remark: High temperatures lead to more creative outputs whereas low temperatures lead to more deterministic ones.



Top-k, top-p

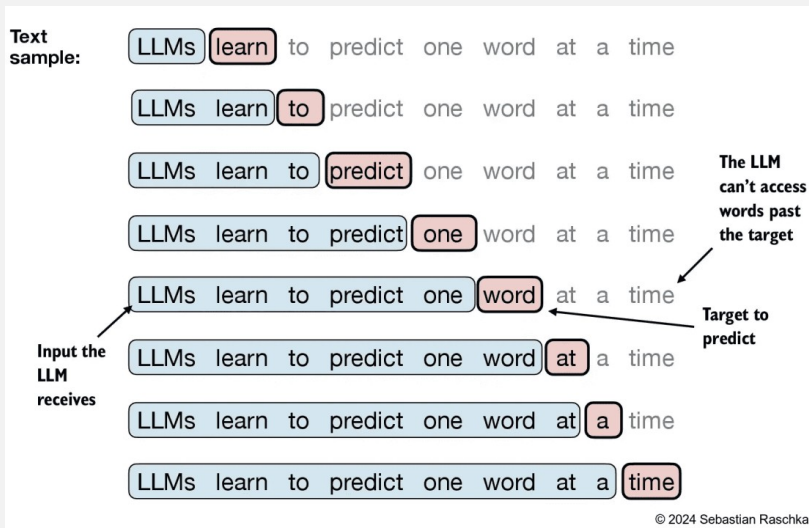


Source:

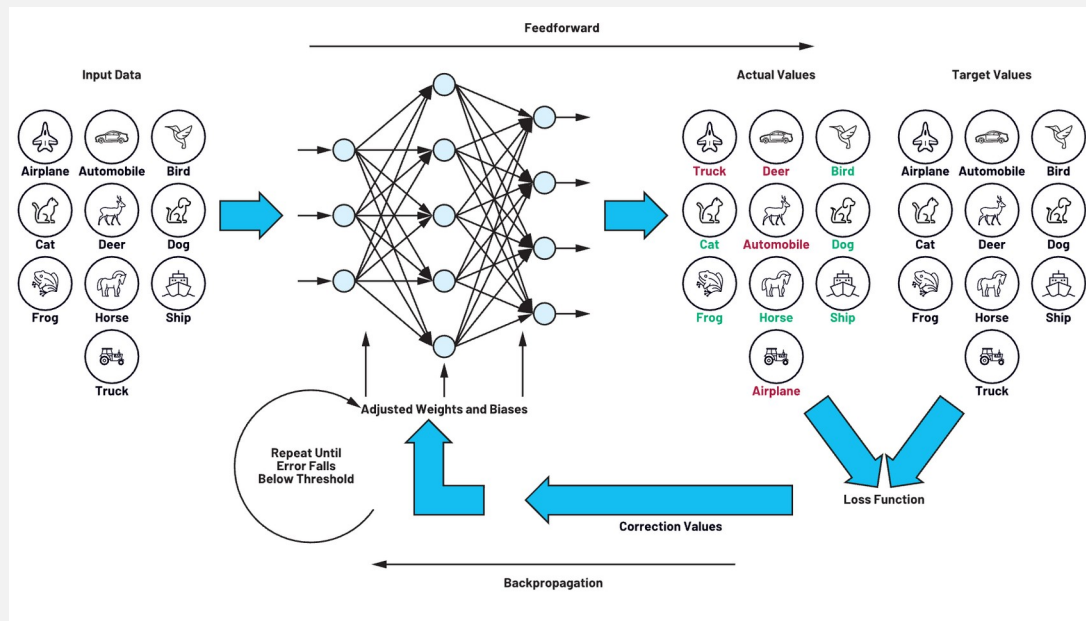
<https://www.geeksforgeeks.org/artificial-intelligence/graph-based-semi-supervised-learning/>

Training: Data sampling (sliding window)

- How to trade off batch size, stride?

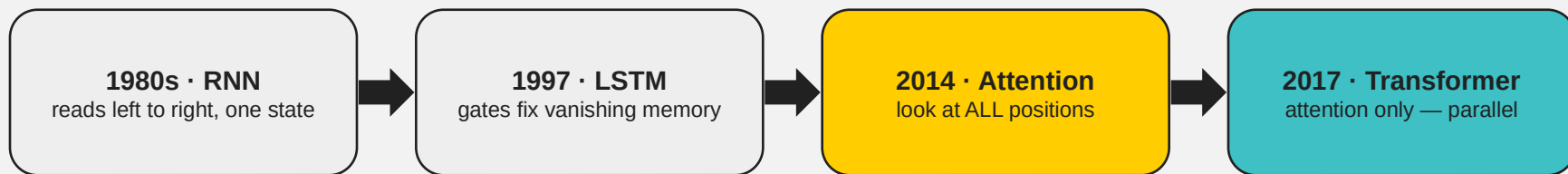


The training loop: forward pass, backward pass



How we got here

Four decades of trying to model sequences: each fix exposed the next limit
Leading question: What are the benefits of Transformer?



Fixed window neural networks

A fixed-window neural Language Model

output distribution

$$\hat{y} = \text{softmax}(Uh + b_2) \in \mathbb{R}^{|V|}$$

hidden layer

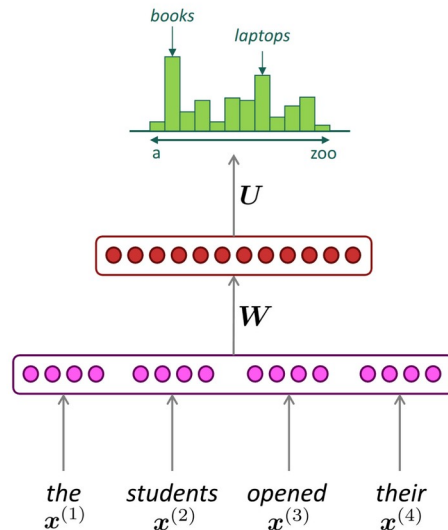
$$h = f(We + b_1)$$

concatenated word embeddings

$$e = [e^{(1)}; e^{(2)}; e^{(3)}; e^{(4)}]$$

words / one-hot vectors

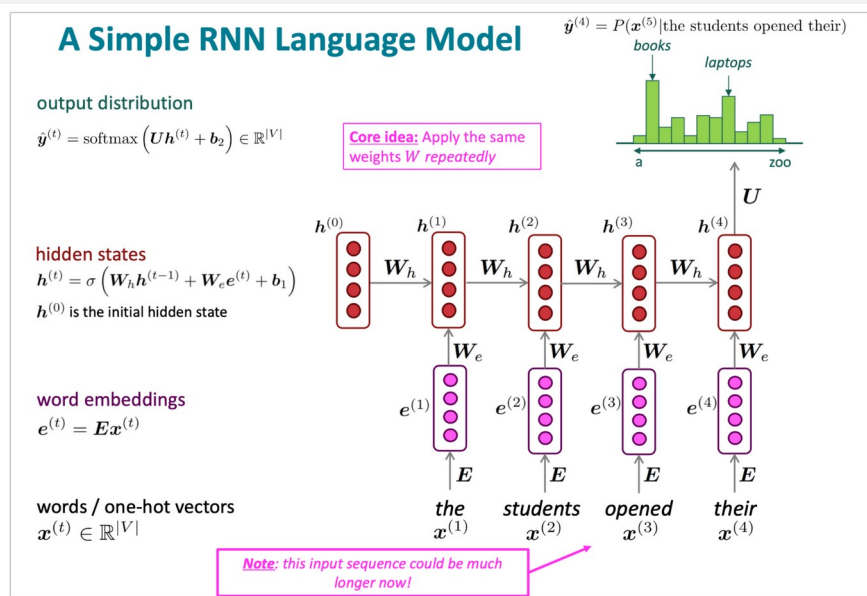
$$x^{(1)}, x^{(2)}, x^{(3)}, x^{(4)}$$



Source: Stanford NLP course.

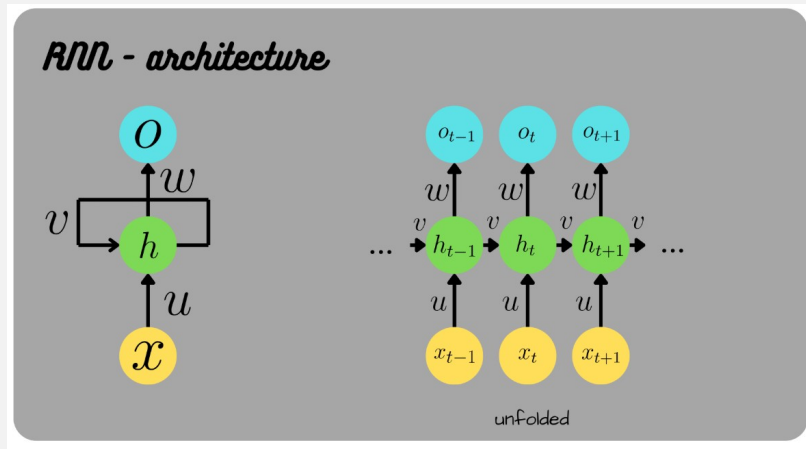
Recurrent Neural Networks

Why is RNN better than fixed window neural networks?



Title: Explaining Recurrent Neural Network (RNN) using an example

Source: [Stanford NLP course](#)



Source:

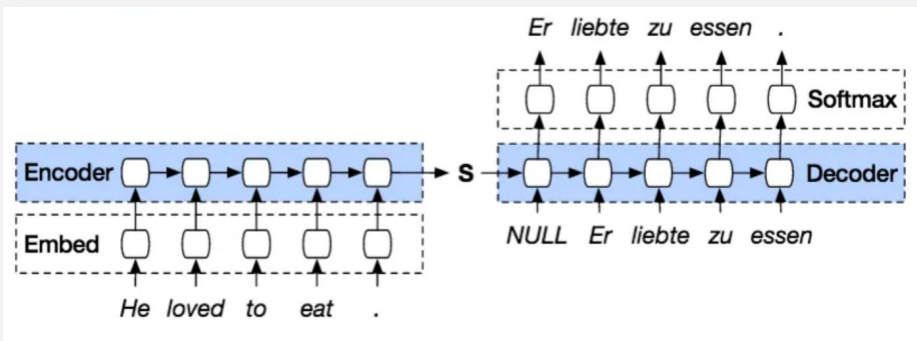
https://datamapu.com/posts/deep_learning/rnn/



Seq2Seq

Neural Machine Translation breakthrough

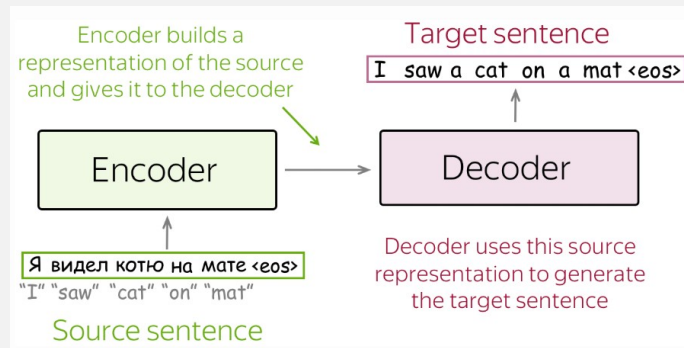
What's the issue in Seq2Seq that's addressed by Transformer?



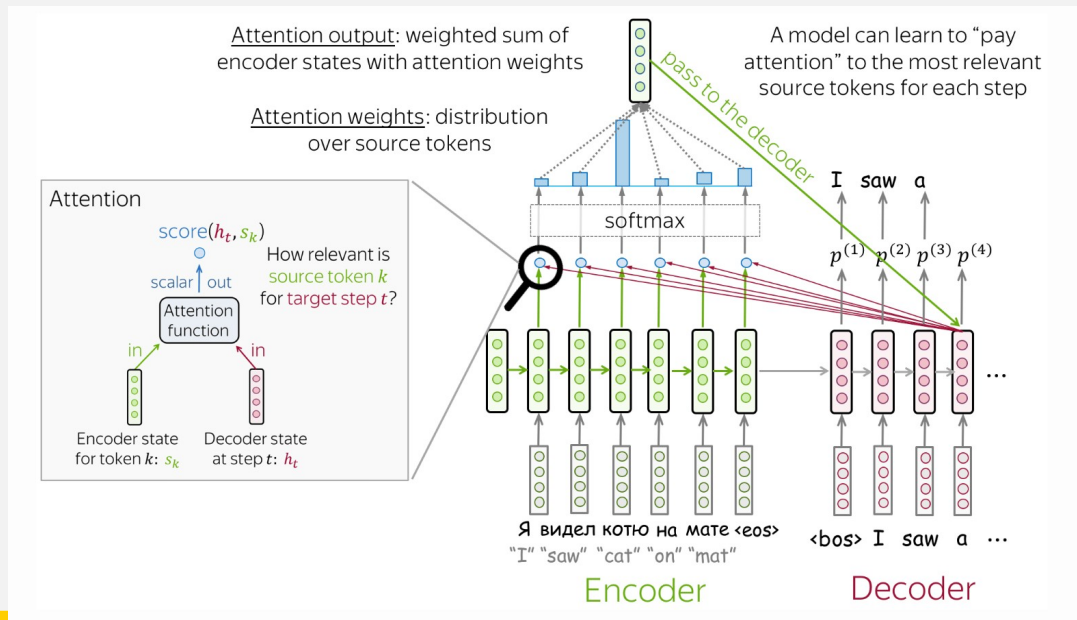
Source:

<https://becominghuman.ai/building-a-seq2seq-transformer-model-for-language-translation-a-comprehensive-guide-875bb7947ee6>

; https://lena-voita.github.io/nlp_course/seq2seq_and_attention.html



Seq2Seq with Attention



“Transformer: Attention is All You Need”

- SOTA at neural machine translation task, then to other tasks...
- “Attention is All You Need”- what does author mean? (next

$$\text{Attention}(q, k, v) = \overbrace{\text{softmax}\left(\frac{qk^T}{\sqrt{d_k}}\right)}^{\text{Attention weights}} \underbrace{v}_{\text{vector dimensionality of K, V}}$$

from to

Each vector receives three representations (“roles”)

$[W_Q] \times \begin{bmatrix} \text{green} \\ \text{green} \\ \text{green} \end{bmatrix} = \begin{bmatrix} \text{blue} \\ \text{green} \\ \text{green} \end{bmatrix}$ Query: vector from which the attention is looking

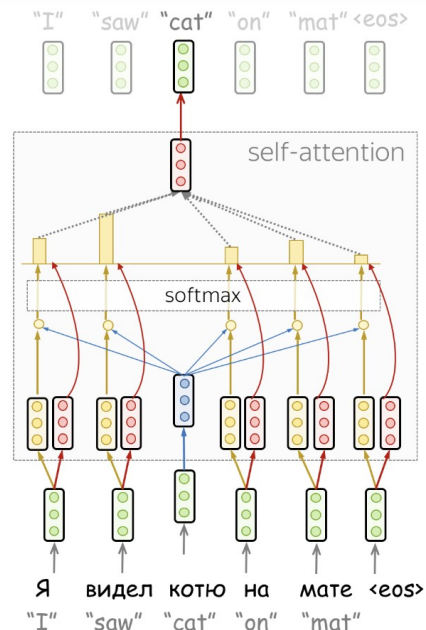
“Hey there, do you have this information?”

$[W_K] \times \begin{bmatrix} \text{green} \\ \text{green} \\ \text{green} \end{bmatrix} = \begin{bmatrix} \text{yellow} \\ \text{yellow} \\ \text{yellow} \end{bmatrix}$ Key: vector at which the query looks to compute weights

“Hi, I have this information – give me a large weight!”

$[W_V] \times \begin{bmatrix} \text{green} \\ \text{green} \\ \text{green} \end{bmatrix} = \begin{bmatrix} \text{red} \\ \text{red} \\ \text{red} \end{bmatrix}$ Value: their weighted sum is attention output

“Here’s the information I have!”



Source: https://lena-voita.github.io/nlp_course/seq2seq_and_attention.html

“Transformer: Attention is All You Need”

	Seq2seq without attention	Seq2seq with attention	Transformer
processing within encoder	RNN/CNN	RNN/CNN	attention
processing within decoder	RNN/CNN	RNN/CNN	attention
decoder-encoder interaction	static fixed- sized vector	attention	attention

Source: https://lena-voita.github.io/nlp_course/seq2seq_and_attention.html

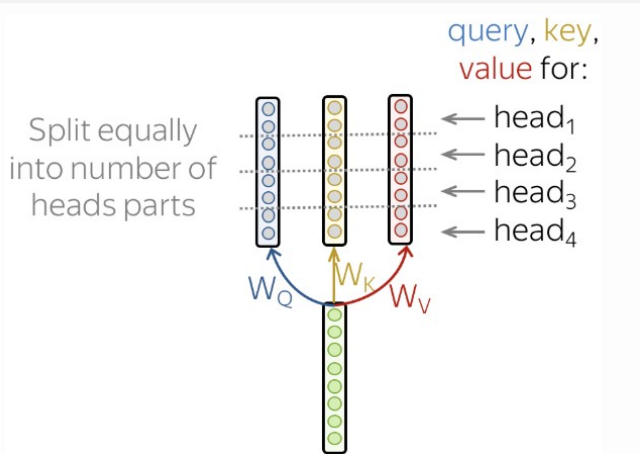


Multi-head attention

$$\text{MultiHead}(Q, K, V) = \text{Concat}(\text{head}_1, \dots, \text{head}_n)W_o,$$

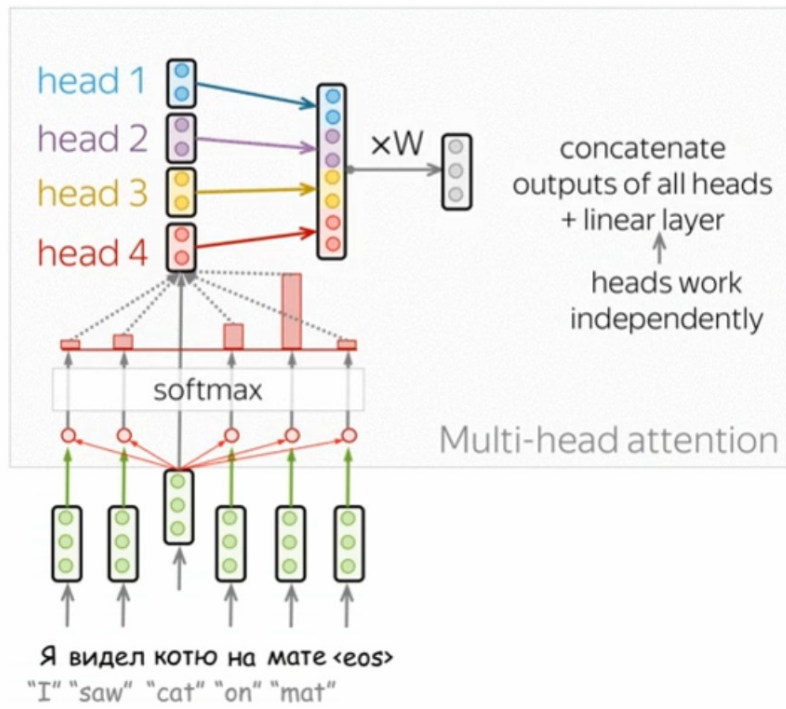
$$\text{head}_i = \text{Attention}(QW_Q^i, KW_K^i, VW_V^i)$$

Why do we need W_o ?



"I" "saw" "cat" "on" "mat" <eos>

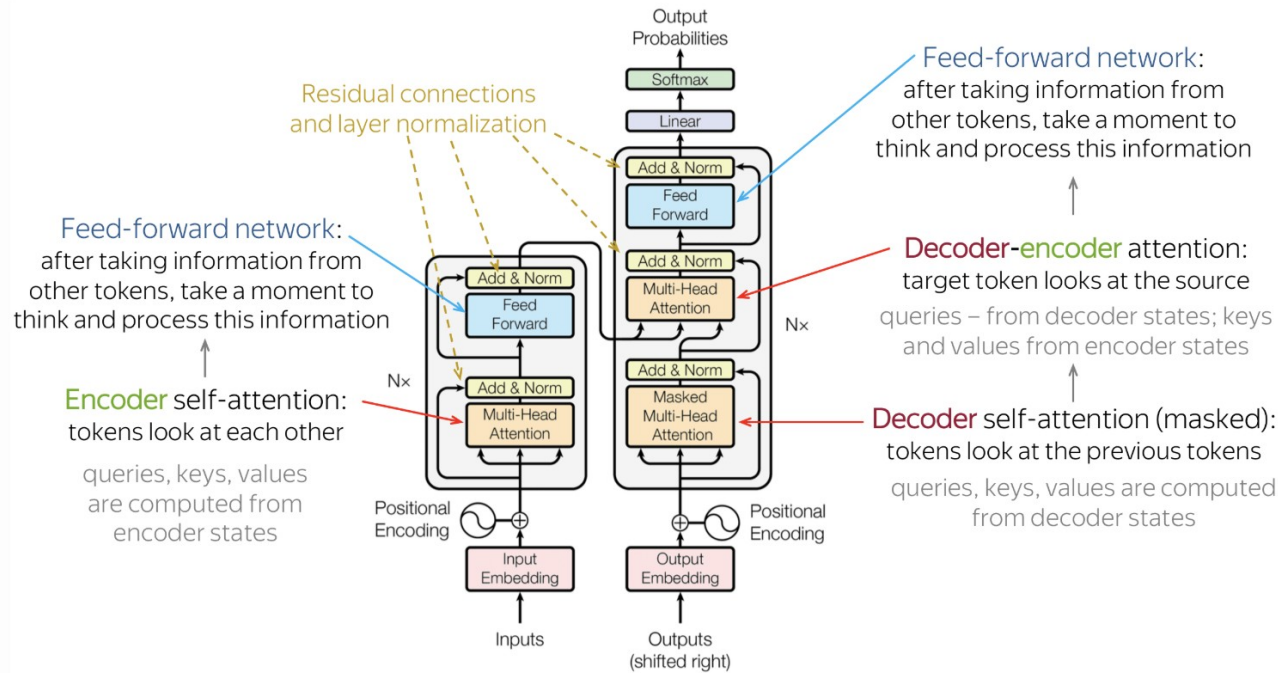
Below the text, each token is represented by a small green rectangle containing three green circles, indicating a vector representation for each word in the sequence.



Transformer Architecture

we have a few things left:

- masked/causal self-attention
- position encoding
- residual connection
- layer normalization
- FFN
- We'll cover them later...



Source: https://lena-voita.github.io/nlp_course/seq2seq_and_attention.html



Example walkthrough

Checkout this visualization! <https://poloclub.github.io/transformer-explainer/>
(another one that visualize nanoGPT: <https://bbycroft.net/llm>)

Please study transformers after class.

- Another visualization youtube:
<https://www.youtube.com/watch?v=wjZofjX0v4M>

Think about a few open questions:

- can we swap self-attention and cross-attention?
- why transformer becomes a mainstream model architecture?
- when to use encoder, or decoder, or encoder-decoder?

LLM Gallery

Compare LLM architecture.

<https://sebastianraschka.com/llm-architecture-gallery/#architecture-gallery>



GPT-2 XL (1.5B)

